INCH-POUND
MIL-DTL-21604/5E
16 February 2017
SUPERSEDING
MIL-DTL-21604/5D
w/Amendment 1
12 May 2016

DETAIL SPECIFICATION SHEET

SWITCH, ROTARY, MULTIPOLE AND SELECTOR, 10 AMPERE, STYLE JR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-DTL-21604.

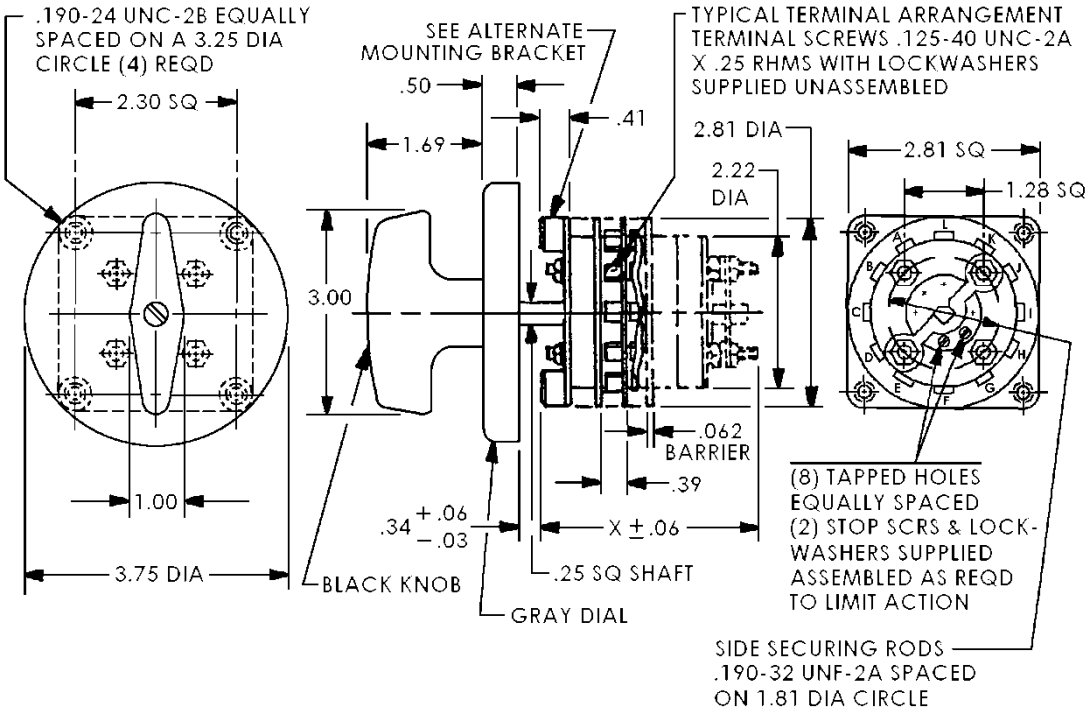
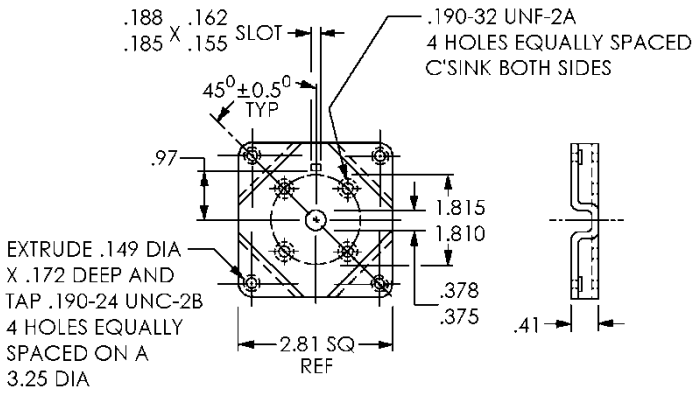
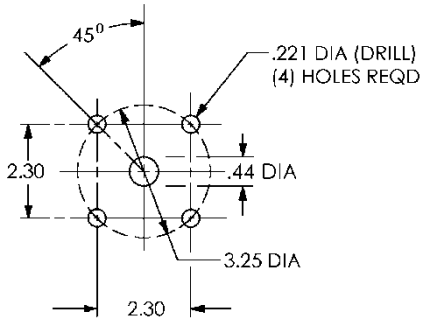


FIGURE 1. Style JR switch.

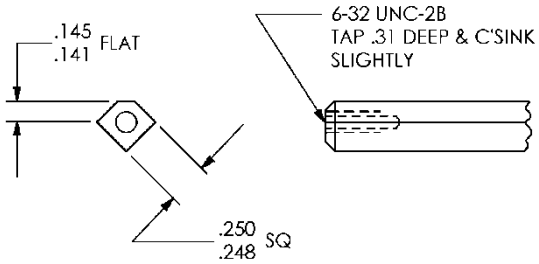
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ALTERNATE MOUNTING BRACKET



PANEL MOUNTING DIMENSIONS
AND REAR SUPPORT



SHAFT

FIGURE 1. Style JR switch - Continued.

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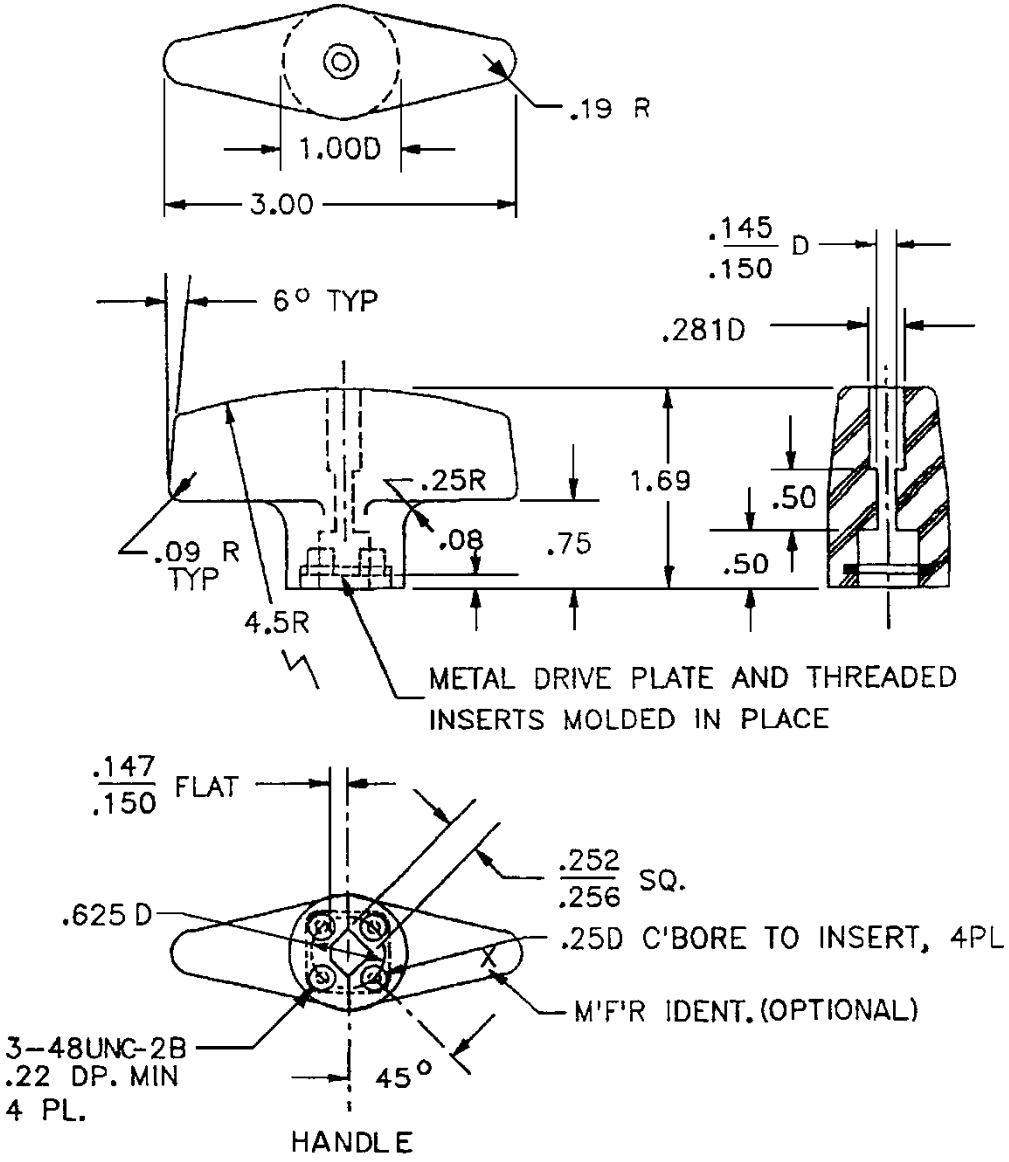


FIGURE 1. Style JR switch - Continued.

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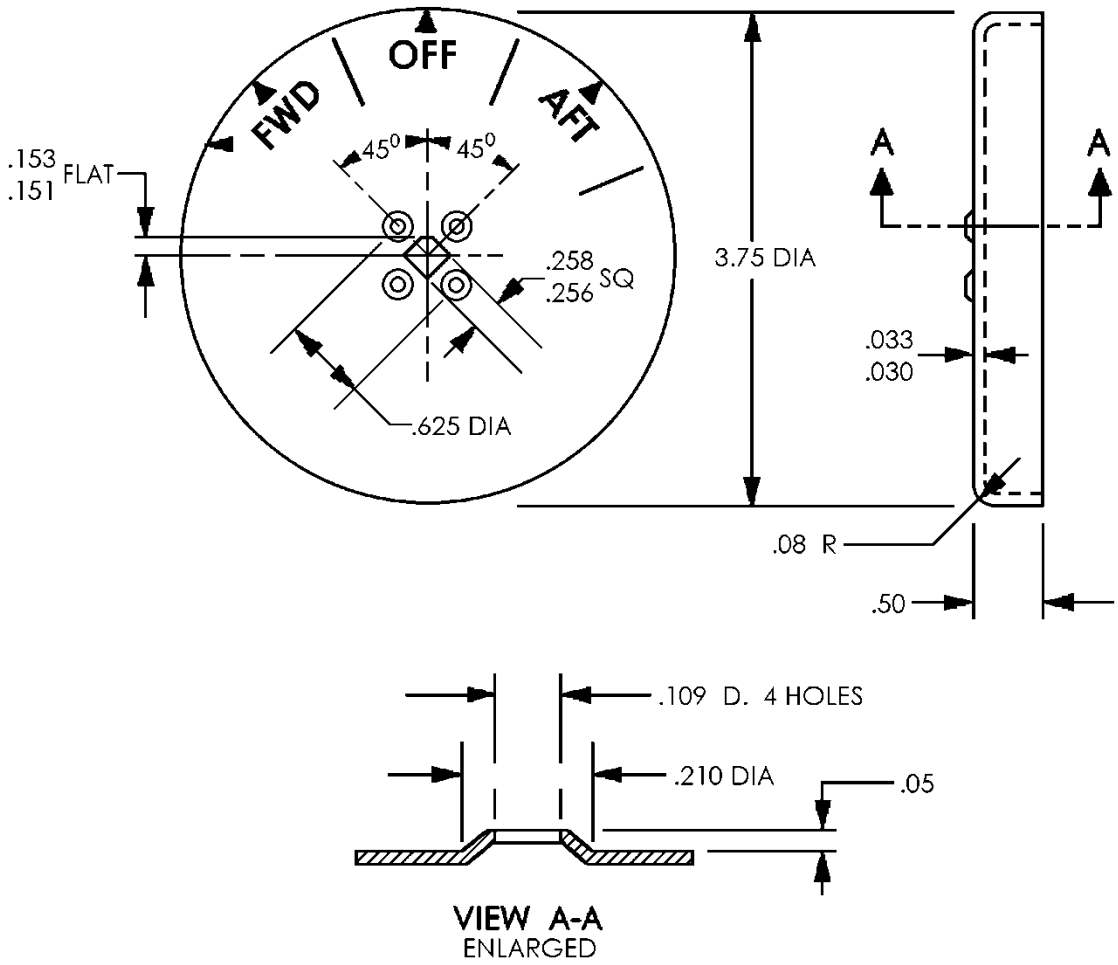


FIGURE 1. Style JR switch - Continued.

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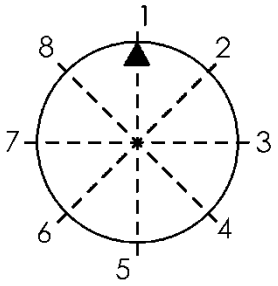
Inch	mm	Inch	mm	Inch	mm
.005	0.13	.185	4.70	.44	11.18
.016	0.41	.188	4.78	.50	12.70
.030	0.76	.19	4.8	.625	15.88
.03	0.8	.190	4.83	.75	19.05
.033	0.84	.210	5.33	.97	24.64
.06	1.52	.22	5.59	1.00	25.40
.062	1.57	.221	5.61	1.28	32.51
.08	2.0	.248	6.30	1.69	42.93
.09	2.3	.25	6.3	1.810	45.97
.109	2.77	.250	6.35	1.81	46.0
.125	3.18	.252	6.40	1.815	46.10
.141	3.58	.256	6.50	2.22	56.39
.145	3.68	.258	6.55	2.30	58.42
.147	3.73	.281	7.14	2.81	71.37
.149	3.78	.31	7.9	3.00	76.20
.150	3.81	.34	8.64	3.25	82.55
.151	3.83	.375	9.52	3.75	95.25
.153	3.89	.378	9.60	4.5	114.3
.155	3.94	.41	10.41	6.0	152.4
.162	4.11				

NOTES:

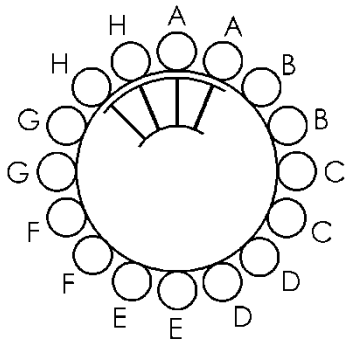
1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .016 (0.41 mm) for two-place decimals and ± 0.005 (0.13 mm) for three-place decimals.
4. On switches of 20 sections or more, the securing rods shall extend .50 inch (12.70 mm) beyond the hex nuts at the rear of the switches to facilitate attachment to rear supports. A hex nut and lockwasher shall be provided for each rod extension.
5. Dial marking, when required, shall be specified in MIL-DTL-21604 (see 6.2).

FIGURE 1. Style JR switch - Continued.

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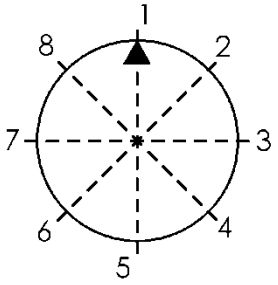


HANDLE POSITIONS

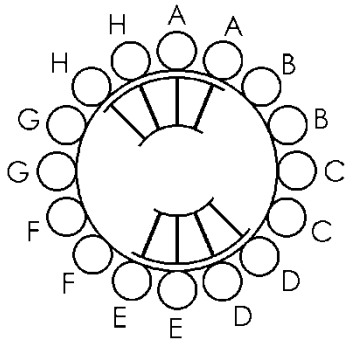


TYPICAL SECTION
SINGLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 1



HANDLE POSITIONS



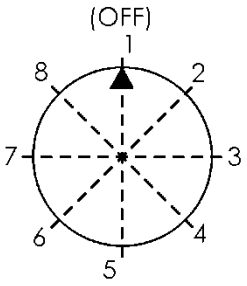
TYPICAL SECTION
DOUBLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 2

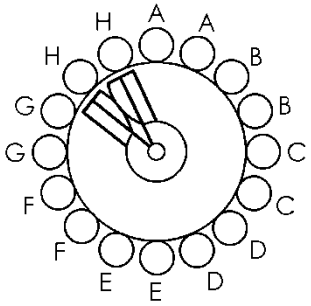
NOTE: Viewed from the handle end.

FIGURE 2. Rotor contact configuration and handle positions.

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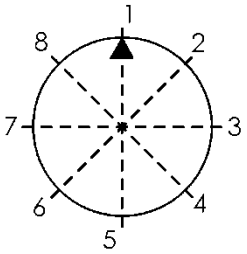


HANDLE POSITIONS

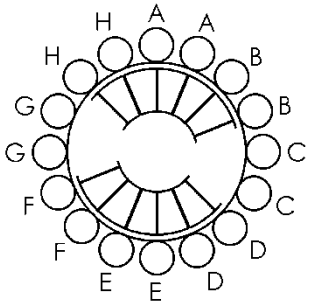


TYPICAL SECTION
SINGLE POLE
SELECTOR

ROTOR DESIGNATION



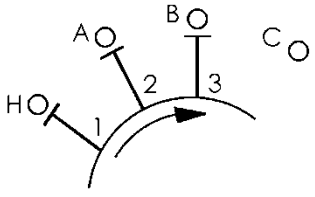
HANDLE POSITIONS



TYPICAL SECTION
DOUBLE POLE
TRIPLE BREAK

ROTOR DESIGNATION 4 1/

1/ WITH ROTOR CONFIGURATION 4, CONTACT H-1 BREAKS BEFORE C-3 MAKES, AND CONTACTS A-1 AND B-2 MAKE BEFORE A-2 AND B-3 BREAK



NOTE: Viewed from handle end.

FIGURE 2. Rotor contact configuration and handle positions - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1, figure 2, and table I.

Angle of throw: 45 degrees.

Switching action: Detent.

Electrical loads:
AC current: 10 amperes.
Voltage: 125 volts (at 60 and 400 Hz, 0.80 inductive power factor).

Temperature rise: 30°C maximum.

Contact material:
Rotor: Silver-overlay phosphor bronze.
Stator: Silver-overlay copper.

Barriers: See figure 1.

Terminal marking: See figure 1 and figure 2.

Stop marking: See figure 1.

Qualification samples: See table II.

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TABLE I. Type designation and switching characteristics.

Type designation		Style	Number of sections	Dimension "X" 2/ ± .062 (1.57)	Approximate weight (ounces)	Maximum rotational torque (inch-pounds)
Component	Rotor designation 1/					
S	1	JR	3	2.97 (75.44)	24	15
S	1	JR	5	3.75 (95.25)	29	20
S	1	JR	6	4.09 (103.89)	31	20
S	1	JR	7	4.47 (113.54)	32	20
S	1	JR	10	5.62 (142.88)	39	30
S	1	JR	15	7.97 (202.44)	48	45
S	1	JR	20	9.94 (252.53)	60	50
S	1	JR	25	11.81 (300.02)	70	55
S	2	JR	2	2.59 (65.79)	23	15
S	2	JR	3	2.97 (75.44)	24	15
S	2	JR	4	3.34 (84.84)	28	20
S	2	JR	5	3.75 (95.25)	29	20
S	2	JR	6	4.09 (103.89)	31	20
S	2	JR	10	5.62 (142.88)	39	30
S	2	JR	11	5.97 (151.64)	40	25
S	2	JR	15	7.97 (202.44)	48	45
S	2	JR	20	9.94 (252.53)	60	50
S	2	JR	25	11.81 (300.02)	70	55
S	3	JR	3	2.97 (75.44)	24	15
S	3	JR	5	3.75 (95.25)	29	20
S	3	JR	7	4.47 (113.54)	32	20
S	3	JR	10	5.62 (142.88)	39	30
S	3	JR	15	7.97 (202.44)	48	45
S	3	JR	20	9.94 (252.53)	60	50
S	3	JR	23	11.37 (288.80)	68	46
S	3	JR	25	11.81 (300.02)	70	55
S	4	JR	5	3.75 (95.25)	29	20
S	4	JR	10	5.62 (142.88)	39	30
S	4	JR	15	7.97 (202.44)	48	45
S	4	JR	20	9.94 (252.53)	60	50
S	4	JR	25	11.81 (300.02)	70	55

1/ For rotor designation, see figure 2.
2/ Dimensions are in inches. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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TABLE II. Qualification test sample.

Sample switch	Test number 1/	Sample switch	Test number
S1JR25	1	S3JR25	4
S1JR25	2	S4JR25	1
S1JR25	3	S4JR25	2
S2JR25	4	S4JR25	3
S2JR25	1	S4JR25	4
S2JR25	2	S4JR25	5
S3JR25	3	S4JR25	6

1/ Test numbers in accordance with table II of MIL-DTL-21604.

Changes from previous issues: The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference Documents:
MIL-DTL-21604

Custodians:
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2017-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

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